ELI LILLY AND COMPANY

Warehouse Capacity Process

	Current Version	Supersedes
Document Number	GSCSOP0125 Rev. 01	SCM-SOP-GSC-00006 Rev. 02

Purpose

This Standard Operating Procedure (SOP) describes the process for assessing and managing warehouse capacity across Lilly manufacturing sites and affiliated organizations.

The procedure outlines the periodic evaluation process used to determine warehouse capacity status for each site and to identify appropriate mitigation measures, such as low-capital initiatives (Local Capital Plan), internal warehouse expansion, or external warehouse utilization.

For external warehouse operations, the applicable business and quality requirements are defined in **Global Quality Standard (GQS-307)** *GMP Service Providers and Consultants* and **LQS-102** *Quality Management of Collaborations with External Parties*. These standards govern the qualification, approval, and ongoing management of third-party (non-Lilly managed) warehouses contracted to store drug products and related materials prior to distribution or use in manufacturing.

When storage capacity constraints are identified, the manufacturing site shall notify **Global Supply Chain (GSC)** and **Global Logistics (GLX)** to initiate a joint review and determine appropriate actions. Approval of any external warehouse project requires review and endorsement by **Global Logistics Senior Leadership** in conjunction with **Manufacturing Leadership**. Identified capacity needs and corresponding mitigation strategies shall be incorporated into the annual **Strategic Plan** and **Business Plan** processes.

This procedure starts with the periodic warehouse capacity evaluation and ends when projects to scale the warehouse capacity to the defined level.

Areas Impacted

- Warehousing & Logistics for Lilly API/SFG/FG Manufacturing Sites
- SFP
- Global Planning

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Tools, Equipment, and Supplies

1.1 NA

2.0 Responsibilities

2.1

3.0 Requirements

3.1 The most recent maximum storage capacity data, along with strategic and business plan information and Warehouse Management System (WMS) data, are available from the manufacturing sites.

3.2 This system pertains to the following types of operations:

- Distribution
- Storage
- Warehousing
- Packaging and Labeling

3.3 This system pertains to the following types of human health products:

- API Starting Material
- Intermediates
- Active Pharmaceutical Ingredients
- Bulk Drug Products
- Drug Products
- Finished Products
- Medical Devices (and device components)
- Drug/Device Combination Products

4.0 Flowchart

4.1 The following figure illustrates the annual process used to evaluate each site's warehouse capacity and, when applicable, to identify and implement actions to address any capacity gaps.

4.2 This process is applied during both the SP and BP cycles.

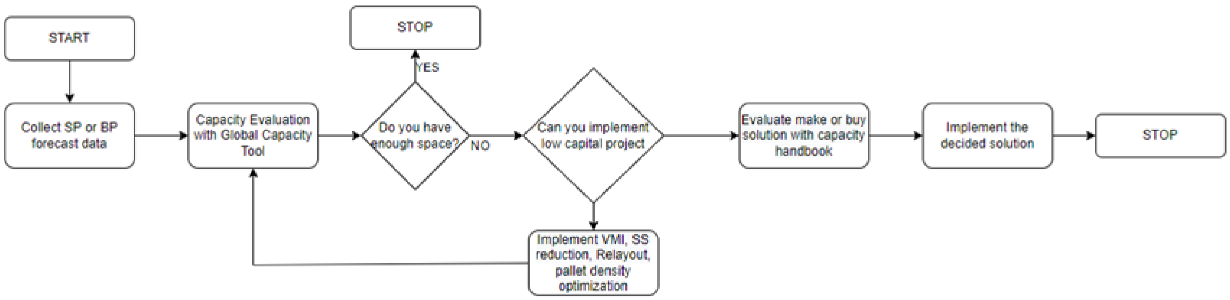

Figure 1. Warehouse Capacity Process

5.0 Procedure

5.1 Governance

5.1.1 Eli Lilly and Company is committed to maintaining sufficient warehouse capacity to store inventory efficiently while supporting operational needs and future growth across all Lilly manufacturing sites. To achieve this, the process outlined in Figure 1 Warehouse Capacity Process, must be followed.

5.1.2 This annual process, led by Global Supply Chain (GSC), begins with the collection of forecasts developed during the Business Plan (BP) and Strategic Plan (SP) volume sessions.

5.1.3 These forecasts are utilized within **the Global Warehouse Capacity Forecast** process to estimate the required warehouse storage capacity levels.

5.1.4 At a minimum, a comprehensive evaluation will be conducted at both the SP and BP stages to assess the global warehouse capacity situation.

5.1.5 The results of this analysis will guide subsequent reviews of capacity and capability across the network.

5.1.6 GSC will engage site logistics subject matter experts (SMEs) at the initiation of the Global Warehouse Capacity Forecast process.

5.1.7 While GSC is responsible for data loading and updates, alignment with each site regarding data inclusions and exceptions is required for every cycle to ensure accuracy and consistency.

5.2 Collect Strategic Plan (SP) and Business Plan (BP) Data

5.2.1 SP and BP data will be processed and provided to Global Logistics by the responsible functions within GSC, specifically Strategic Facility Planning (SFP) and/or Global Planning.

5.2.2 Global Logistics will be responsible for converting this data into the format required by the **Warehouse Capacity Tool (WCT)** or alternatives to enable accurate processing and analysis of the results

5.3 Capacity Evaluation

5.3.1 GLX is responsible for determining the warehouse occupancy level for each site across the defined analysis period (SP or BP horizon) and calculate the initial occupancy level to 85% utilization

NOTE: Monitoring and adjusting a site's utilization is a shared responsibility between the site and GLX

5.3.2 The Warehouse Capacity Tool (WCT) used for this evaluation integrates multiple data sources, including:

- Strategic Plan or Business Plan Data
- Key historical information from the Warehouse Management System (WMS),
- Internal and external warehouse capacity data specific to each manufacturing site

5.3.3 Historical Warehouse Management System (WMS) data includes the calculated average warehouse occupancy rate, derived from monthly inventory snapshots for the 12-month period preceding the analysis.

5.3.4 In addition to material movement data, the dataset integrates volume projections and demand forecasts aligned with GS&OP processes.

5.3.5 These inputs enable accurate capacity planning, inventory optimization, and scenario modeling to support resiliency and operational continuity across the supply chain network.

5.4 Space Evaluation

5.4.1 The results of the capacity analysis provide detailed occupancy data for each warehouse area recorded in the WMS at every production site.

5.4.2 This occupancy data, expressed in the number of pallets or handling units (HU), is compared against the defined capacity for each area.

5.4.3 Any areas identified as exceeding or nearing capacity thresholds are flagged for further evaluation.

5.4.4 Upon completion of the site capacity analysis, the local manufacturing site SME and GSC representative will meet to align upon the following prioritized points:

5.4.5 **Space Evaluation Priorities**

- **Baseline Validation:** Confirm that the baseline accurately reflects the prior period's warehouse occupancy level based on WMS inventory data.
- **Site Optimization:** Evaluate on-site storage availability and constraints to identify materials that can be reallocated to other storage locations with available capacity (i.e., storage balancing).
- **Forecast Alignment:** Ensure alignment between site leadership and GSC regarding forecasted occupancy levels.

5.4.6 If occupancy levels are determined to be within acceptable limits as defined per **Manufacturing Warehouse Strategy for API/Drug Substance and Drug Product – STG-08601**, the evaluation process concludes.

5.4.7 If additional warehouse capacity is required, Global Logistics (GLX) and Site Manufacturing Leadership will jointly develop a plan to address the expansion needs.

5.4.8 If the site elects not to prioritize expansion, Attachment D – Exception to Business Process Request Form must be completed by the manufacturing site.

5.4.9 The form must be signed by the Site Head and Global Logistics Associate Vice President (AVP) to document that the analysis was performed and the decision not to proceed was formally approved.

5.5 Low Capital Project

5.5.1 Following the completion and alignment of the Space Evaluation, the first potential solution is an internal expansion implemented through a low-capital project.

5.5.2 A low-capital project shall be managed as a formal project and executed within the framework of the local capital plan process using available site resources.

5.5.3 Low Capital Project Examples:

- Installation of new freezers for APIs
- Shelving Remodulation
- Re-layout of manual warehouses
- Minor adjustments to inventories or site buffers

5.5.4 In cases where a low-capital project does not sufficiently address the site's internal expansion needs, a **"make versus buy"** evaluation will be conducted to determine the most effective solution.

5.6 Evaluation Make versus Buy

5.6.1 Once the need for additional space has been confirmed and is associated with the addition of new production lines and/or a site expansion, a formal project shall be established and managed in accordance with Project Management Office (PMO) principles and oversight.

5.6.2 GLX is responsible for recommending the appropriate course of action to the site based on the following criteria.

5.7 Evaluated Internal & External Warehouse Expansion Parameters

- **Duration and go-live requirements** (i.e., the timeline for space availability)
- **Material matrix considerations**
- **Planned days on hand (DOH)** for inventory availability within the internal warehouse for materials eligible for external storage, as defined by the material matrix.

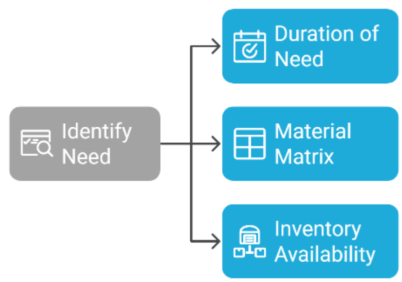

Figure 2. Warehouse Expansion Parameters

5.8 Risk Assessment for External Warehouse Operations and 3PL Selection

5.8.1 External warehouse operations introduce risk factors such as natural hazards, cost variability, contract terms, and distance to manufacturing sites.

5.8.2 When engaging a third-party logistics (3PL) provider, additional considerations include specification control, service reliability, and compliance with engineering standards (e.g., FM Global).

5.8.3 The selection of a 3PL location requires a holistic review of these risk elements, project timeline, warehouse size, and associated revenue protection, alongside volume projections and operational requirements.

5.8.4 Final decisions are governed by the Global Supply Chain (GSC) Resilience Team in collaboration with the Global Risk Team to ensure continuity, safety, and strategic alignment.

5.9 Duration and Go-Live

5.9.1 The duration of the required capacity is a critical parameter in the decision-making process.

(1) For temporary or short-term needs, the preferred approach is typically to utilize external warehouse space, even if this deviates from the parameters defined in the material matrix. This approach is justified, as investing significant capital expenditure (CAPEX) for space that will only be used for a limited period does not provide a sustainable benefit to the company.

(2) The second key parameter is the material matrix (refer to the Material Matrix in the Manufacturing Warehouse Strategy for API/Drug Substance & Drug Product - STG-08601), which serves as a primary guide for determining whether internal or external storage is required under standard operating conditions.

(3) The third parameter concerns the potential impact on the internal buffer at the site. This buffer typically ranges from three to five (3 to 5) days for raw materials. However, during an expansion, it may decrease below three (3) days, in which case an internal adjustment must be implemented. When the buffer drops below this threshold, it indicates that the site's internal warehouse capacity is at or near its maximum limit. This situation generally necessitates a corresponding adjustment to external capacity in alignment with the hybrid storage strategy, which balances internal and external space utilization.

5.9.2 After all these elements have been evaluated, GLX will issue a recommendation. At the global level, this evaluation is recommended to the GLX Lead Team.

5.9.3 If GLX Lead Team "agrees"/ approves the recommendation, then the recommendation, is then brought to Manufacturing Leadership Team (MLT) for final sourcing decision approval.

5.10 MLT Sourcing Decisions Process Overview

5.10.1 Global manufacturing operations require consistent evaluation of sourcing options to balance capacity, quality, cost, and regulatory requirements.

5.10.2 To achieve this, Lilly employs a two-gate decision framework that provides governance and transparency in determining where and how manufacturing activities occur. This structured approach supports strategic alignment, cost optimization, and risk mitigation.

5.10.3 The process begins with Gate 1 (see Figure 3) which evaluates internal and external manufacturing options through a comprehensive business case analysis. This includes comparing cost, capability, and product-specific data to define boundary conditions and assumptions.

5.10.4 Internal recommendations require Manufacturing Leadership Team (MLT) approval to insource, while external recommendations proceed to Gate 2.

5.10.5 Gate 2 finalizes the sourcing decision by selecting the specific Lilly site or external partner and negotiating supply parameters.

5.10.6 It also addresses implementation requirements such as expenses, timelines, and ongoing operational oversight.

5.10.7 This two-step approach ensures informed decisions that support business objectives, compliance, and long-term sustainability.

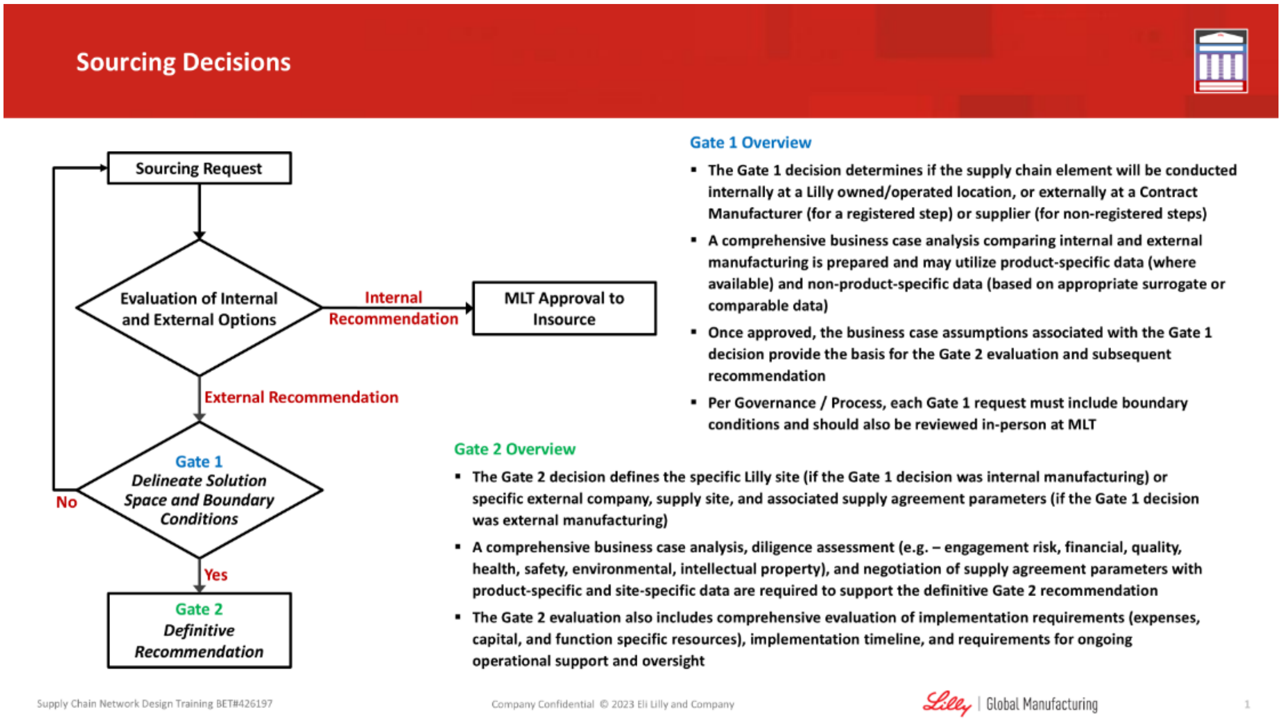

Figure 3. Sourcing Decisions

5.11 Implement the Solution

5.11.1 The implementation phase for most capital-intensive projects is typically the longest stage of execution. Prior to initiation, all necessary requirements must be clearly defined and documented, including the number of pallet positions required, storage condition specifications, desired automation levels, fire protection systems, transportation needs, and throughput requirements.

5.11.2 If the project involves an internal expansion, two possible pathways apply:

(1) Global capital project: Follow the operating procedures established by the Global Facilities Delivery (GFD) organization.

(2) Local capital project: Follow the applicable local site procedures.

5.11.3 For external warehouse expansions, Global Logistics and Global Procurement will initiate a Request for Information/Proposal (RFI/P) process to gather the necessary information from existing external logistics partners as well as other qualified market providers.

5.11.4 The evaluation will be conducted in alignment with Global Procurement’s External Vendor Strategy.

5.11.5 The assigned project team will assess and rank potential partners, recommending the most suitable candidate for approval.

5.11.6 This evaluation will consider the Total Cost of Acquisition (TCA) over the expected duration of the engagement.

5.11.7 Based on the TCA results, different approval levels may be required (refer to the Procurement Playbook for the appropriate approval thresholds).

5.11.8 Once all required approvals have been obtained, Global Procurement will formally notify the selected partner and manage the contractual agreement process.

5.11.9 Concurrently, the organization and/or local site utilizing the expanded warehouse will be responsible for initiating change control, documenting, and executing all required activities to ensure readiness and compliance for operational use of the new space.

6.0 Terms and Acronyms

6.1 **AGV** – Automated Guided Vehicle

6.2 **AMR –** Autonomous Mobile Robots

6.3 **API** – Active Product Ingredient

6.4 **ASRS** – Automated Storage and Retrieval System

6.5 **BP** – Business Plan

6.6 **CDA** – Confidential Disclosure Agreement

6.7 **DC** – Distribution Center

6.8 **DOH** – Days on Hand

6.9 **DDMRP –** Demand Driven Material Requirements Planning

6.10 **FG** – Finished Goods

6.11 **FRAP –** Financial Responsibility and Authorization Policy

6.12 **GQS** – Global Quality Standards

6.13 **GSC** – Global Supply Chain

6.14 **HSE** – Health and Safety/ Environmental

6.15 **HU** – Handling Unit (E.g. Pallet, Bottles, Trays, Container…)

6.16 **LSP –** Logistic Service Provider, also named Third Party Logistic

6.17 **MSDS –** Material Safety Data Sheet

6.18 **QA** – Quality Assurance

6.19 **RFP –** Request for Proposal

6.20 **SC** – Supply Chain

6.21 **SFG –** Semi-Finished Goods

6.22 **SP –** Strategic Plan

6.23 **3PL –** Third Party Logistic

6.24 **WIP –** Work In Process

7.0 Related Documents

	Document Number	Document Title
7.1	EBP-0130	Warehouse Facility, Engineering Best Practices
7.2	GQS-202	Facilities
7.3	GQS-307	GMP Service Providers and Consultants
7.4	LQS-102	Quality Management of Collaborations with External Parties
7.5	GLX-RES-0003	Pharmaceutical Warehouse Capacity Management Handbook
7.6	STG-08601	Manufacturing Warehouse Strategy for API/Drug Substance and Drug Product
7.7	PR-9487940	Procurement Playbook
7.8	GSCSOP0125-TOOL-01	External Warehouse Implementation Process
7.9	GSCSOP0125-TOOL-02	Warehouse Dimensioning
7.10	GSCSOP0125-TOOL-03	Warehouse Suggested Related Metrics
7.11	GSCSOP0125-FORM-01	Exception to Business Process Request Form

8.0 Reasons for Revision

8.1 Separation of warehouse strategies and warehouse capacity processes throughout.

8.2 Update of title to remove “Strategy”.

8.3 Updated to new naming convention and template per GSCSOP0001.

8.4 Created Tools and Forms from previous attachments within SOP.

Document Approval Signatures

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